

5.1 Approximating Areas

In Chapters 1 – 4 we have looked at the application of limits as it related to tangent lines and their slopes. In Chapter 5 turn our attention to a seeming unrelated question of using limits to help determine areas.

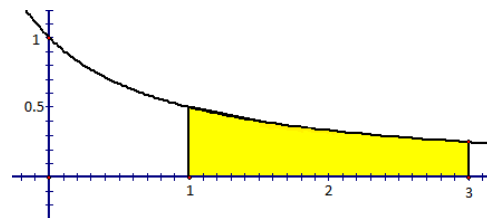
General Idea (Part I):

A car is traveling at a constant rate of 50mph for a 2-hour period. Find the total change in position. Represent this quantity as an area on a graph.

Note: When the rate of change of a quantity is positive, the total change in that quantity on some interval can be represented by the area between a function and the horizontal axis on that interval.

General Idea (Part II):

An object travels at $v(t) = \frac{1}{t+1} \text{ ft/sec}$ between the times $t = 1$ and $t = 3$ seconds. Estimate the total change in position of the object during this time.

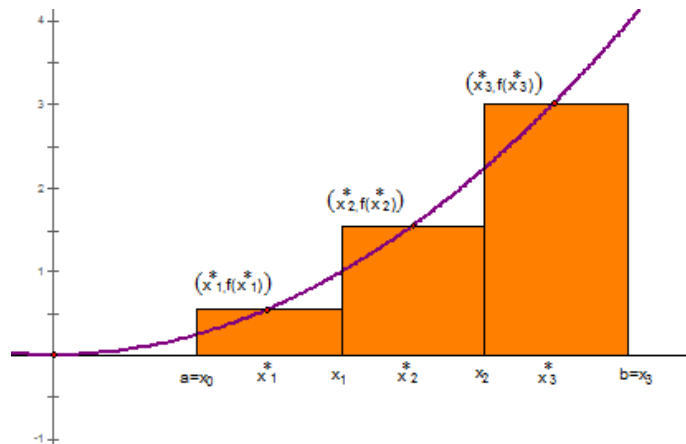


Exploration: The area A of the region S that lies under the graph of the continuous function f can be approximated by using sum of areas of n approximating rectangles.

If

- each rectangle has an equal width
- each rectangle has height

Then the area of each rectangle will be given by the formula



Therefore the approximation of the area will be given by

Definition: The net area (also called signed area) of a region that lies between the x -axis and the graph of a continuous function f may be approximated with a Riemann sum which is a sum of the form

Common Riemann Sums:

- A *Left Riemann Sum* has the top left corner of each rectangle on the function.
- A *Right Riemann Sum* has the top right corner of each rectangle on the function.
- A *Midpoint Riemann Sum* has the midpoint of the top side of each rectangle on the function.

Example 1: Let A be the area of the region that lies above the x -axis and under the graph of $f(x) = x^2$ between $x = 0$ & $x = 2$.

- Estimate the area by using a Left Riemann Sum with 4 rectangles.

b) Estimate the area by using a Right Riemann Sum with 4 rectangles.

c) Estimate the area by using Midpoint Riemann Sum with 8 rectangles.

Note: If f takes on both positive and negative values, then the rectangles above the x -axis have positive area and the rectangles below the x -axis have a negative area.

Example 2: Let A be the area of the region that lies between the x -axis and $f(x) = x^3 - 4x$ between $x = -1$ & $x = 2$.

a) Estimate the area by using a Riemann sum with left endpoints and 3 rectangles.

b) Estimate the area by using a Riemann sum with right endpoints and 6 rectangles.

Example 3: Suppose the odometer on our car is broken and we want to estimate the distance driven over a 30 minute time interval. We take the speedometer readings every 5 minutes $\left(\frac{1}{12} hr\right)$ and record them in the following table:

Time (hours)	0	1/12	2/12	3/12	4/12	5/12	6/12
Velocity(mi/h)	17	21	24	29	32	31	28

a) Estimate the distance traveled using left endpoints.

b) Estimate the distance traveled using right endpoints.